

BRUCE DAVIS

AI SYSTEMS ENGINEER · AGENTIC AI · IAM · SECURITY

TARGET ROLE: AI Systems Engineer · Forward Deployed AI Engineer · Applied AI Engineer · Senior AI Engineer – Agentic Systems
Open to full-time, contract, and project work. Remote-first.

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Portfolio

I build agentic AI systems — MCP servers, ReAct agent loops, multi-agent orchestration, RAG pipelines. The fifteen years before that were in identity and security: **Non-Human Identity (NHI) governance at Microsoft**, HIPAA-compliant IAM for 16,000+ users at Veradigm, payment infrastructure for SWIFT/CHIPS/Fedwire at Wells Fargo. Deploying an AI agent safely is mostly an identity and authorization problem, which is the work I already know. PMP-certified.

SELECTED PROJECTS – AGENTIC AI, IAM, SECURITY

AUTONOMOUS SECURITY TRIAGE AGENT

Human-in-the-loop authorization on a ReAct agent loop

ReAct agent loop on Gemini 2.5 Flash — reads logs, enriches threat intelligence, proposes containment actions through function-calling tools. Destructive actions pause for explicit human approval before execution. An autonomy dial scopes operational reach. Built for security teams that need AI assistance without surrendering control. Python · Streamlit · Gemini 2.5 Flash · google-genai SDK · ReAct · Function Calling · HITL Architecture.

MULTI-AGENT INCIDENT INVESTIGATION SYSTEM

Non-Human Identity (NHI) governance applied to a six-agent system

Six AI agents investigate a live security incident — each with its own identity, permission level, token budget, and accountable owner. When an agent needs elevated permissions, the system stops and requests human authorization before granting them. Non-Human Identity governance applied directly to AI agents. Python · React · TypeScript · Multi-agent Architecture · NHI · Generative UI.

DOCUMENT PROCESSING PIPELINE

HIPAA-ready extraction, PII masking, validation, and audit trail

An MCP tool chain handles extraction, PII masking and deidentification, structured JSON output against a validated schema, human-in-the-loop review for edge cases, and token-cost tracking. Built on Claude; runs on GPT-4, Gemini, or any other tool-calling model. Made for HIPAA and other compliance-sensitive workflows that need a full audit trail. MCP SDK · Node · Python · ajv · JSON Schema · PII Governance.

AI INTEGRATION FOR LINEAR

Production MCP server with threat model and evaluation harness

Six opinionated MCP tools for engineering team operations — notification triage, issue scoping, velocity diagnosis, status reporting. Includes a 30-case evaluation harness, a documented threat model, and per-tool cost analysis. Open source. TypeScript · @modelcontextprotocol/sdk · Linear GraphQL API · Zod · Vitest.

CONTENT & RESEARCH AUTOMATION

Custom MCP server — Notion-backed retrieval, tone analysis, scaffolding, vault capture

Custom MCP server with five tools for a content and research workflow — Notion-backed idea retrieval, tone analysis, page scaffolding, vault capture. I use it day to day. @modelcontextprotocol/sdk · Notion API · Zod · Node.

FILM PRE-PRODUCTION AGENT

One agent architecture, validated across a second domain

The same ReAct agent loop as the Autonomous Security Triage Agent, pointed at film pre-production — script breakdown, VFX complexity scoring, casting archetypes, producer-approval gate before lock. The same architecture works in a second domain. Python · Streamlit · Gemini 2.5 Flash · google-genai SDK · ReAct · Function Calling.

WORK

Business Execution Consultant — Solution Design

Nov 2024–Apr 2025

Wells Fargo · Contract

- Mapped network dependencies and ran gap analysis across SWIFT, CHIPS, and Fedwire payment infrastructure — documented system flows that hadn't existed on paper, accelerated regulatory approval, and cut design rework by 25%.
- Designed NIST-compliant documentation architecture for network security controls; became the cross-team reference layer for incident response.
- Decomposed executive payment-rail objectives into detailed execution plans for ISO 20022 interoperability and Federal Reserve compliance.

Cloud Solutions Architect — AI

May–Nov 2024

Microsoft · Contract

Forward-deployed Cloud Solutions Architect aligned to Microsoft's AI Factory — embedded with enterprise customers to design, ship, and deploy Azure AI solutions.

- Built Python automation on the Microsoft Graph API for Entra ID Governance — extracted and normalized identity data into executive-ready dashboards and architecture documentation.
- Architected and shipped a Generative AI workflow MVP on Microsoft Copilot — accelerated knowledge transfer and standardized requirements gathering across global delivery teams.
- Translated Non-Human Identity (NHI) governance capabilities into customer-facing solution designs.

Senior Security Engineer

Apr–Sept 2023

Starbucks · Contract

- Engineered PowerShell automation across 60+ domains for identity lifecycle and access control remediation — shipped scripts and runbooks at the engineering layer.
- Closed 50+ vulnerabilities per sprint through automation and scripted remediation rather than manual triage.
- Hardened identity and access controls across distributed teams — configured authentication protocols, owned security testing, and ran defect remediation end to end.

Project Manager — Privileged Access Management (PAM)

Jul 2021–Nov 2022

Bank of America · Contract

- Led zero-downtime IAM and infrastructure modernization for 12+ teams and 60+ engineers across Oracle, CyberArk, Azure, Middleware, and API platforms — coordinated dependency mapping, change validation, and stakeholder alignment.
- Directed HashiCorp Vault enterprise rollout and service account password rotation automation — integration planning, stakeholder communication, and change management.
- Owned ServiceNow dashboards as operational source of truth for change approvals, release readiness, and migration tracking; ensured all initiatives met stringent audit requirements.
- Authored runbooks, migration plans, and governance documentation for senior leadership; identified delivery risks through deep-dive requirements workshops.

Senior IAM Security Engineer

Jan 2019–Jun 2021

Veradigm · Healthcare

- Architected and shipped an IAM platform for 16,000+ users in a HIPAA-regulated healthcare environment — Entra ID, MIM, and CyberArk integrated end to end.
- Built Power Platform dashboards that surfaced compliance state across the organization; moved audit compliance from 65% to 95% over the engagement.
- Designed and executed a multi-year IAM strategy in a regulated healthcare environment, overseeing technology selection and solution architecture.

Earlier: Data Quality Analyst · Wells Fargo (2017–2018) · IAM Consultant · Cognizant (2016–2017) · Support Escalation Engineer · Microsoft (2012–2015)

STACK

FOUNDATION & SMALL MODELS: OpenAI · Google · Anthropic · Meta Llama · Gemini · Mistral · Hugging Face · Open-source models

ORCHESTRATION & FRAMEWORKS: MCP · LangChain · LlamaIndex · Vertex AI · Agent platforms · Workflow orchestration · ReAct · Function Calling · Multi-agent Architecture · RAG · Human-in-the-Loop (HITL) · Anthropic SDK · OpenAI SDK · google-genai SDK · Vercel AI SDK

APPLICATION & UI: Claude · ChatGPT · Gemini · Microsoft Copilot · Notion AI

CLOUD & INFRASTRUCTURE: Microsoft Azure · Google Cloud Platform · Vercel

IDENTITY & SECURITY: Non-Human Identity (NHI) · Entra ID Governance · CyberArk · Microsoft Identity Manager (MIM) · HashiCorp Vault · Zero Trust · HIPAA · NIST · ISO 20022 · SWIFT · CHIPS · Fedwire · Threat Modeling · ServiceNow

ENGINEERING: Python · TypeScript · JavaScript · Node.js · React · Next.js · Streamlit · PowerShell · SQL · GraphQL · Zod · Vitest · Git · Vercel · Tailwind

DELIVERY: PMP · Scrum (PSM I) · Microsoft AI Factory · Forward Deployed Engineering · Audit-Ready Documentation

CERTIFICATIONS & EDUCATION

Google UX Design Professional Certificate

Google (via Coursera) · 2023

Certified in Cybersecurity (CC)

ISC2

Project Management Professional (PMP)

Project Management Institute

Professional Scrum Master I (PSM I)

Scrum.org

M.S., Computer Graphics Technology

Purdue University

Thesis: An Exploration of the Role of Interactivity in Video Game Based-Learning

B.S., Computer Science

Lincoln University

Information Systems

Carnegie Mellon University